

PROJECT CHARTER

UC11: Enterprise Pharmaceutical Data Platform (EPDP)

Building a Unified Pharmaceutical Manufacturing & Clinical Analytics Platform
Sprint 0 & Sprint 1 (DA-1 Submission)

Document Information

Organization	ABC Pharmaceuticals Ltd.
Project Code	UC11
Project Phase	Sprint 0 & Sprint 1
Methodology	Agile Scrum
Version	1.0
Prepared By	Alfred, Sanika, Thomas

1. Project Overview

The Enterprise Pharmaceutical Data Platform (EPDP) is an enterprise data engineering project that integrates pharmaceutical manufacturing, laboratory, quality assurance, inventory, clinical research, regulatory, and IoT datasets into a centralized PostgreSQL Data Warehouse using Pentaho ETL pipelines and Python-based data profiling.

2. Business Need

ABC Pharmaceuticals Ltd. operates multiple independent enterprise systems. The project addresses data integration, product traceability, manufacturing quality monitoring, laboratory data consistency, inventory management, and regulatory reporting.

3. Project Objectives

- Identify pharmaceutical data sources.
- Build Pentaho ETL pipelines for CSV, Excel, JSON, XML, and SQL datasets.
- Clean, validate, and standardize pharmaceutical data.
- Load raw data into PostgreSQL staging tables.
- Prepare data for warehouse development and analytics.
- Maintain GitHub repository for collaborative development.

4. Project Scope

IN SCOPE (Sprint 0 & Sprint 1)

Sprint 0:

- Business Requirement Document (BRD)
- Project Charter
- Stakeholder Identification
- High-Level Solution Architecture
- GitHub Repository Setup
- Product Backlog
- Sprint Backlog

Sprint 1:

- Source Inventory
- Data Dictionary
- Pentaho ETL Pipelines
- CSV, Excel, JSON, XML, SQL Data Ingestion
- PostgreSQL Staging Tables
- ETL Logging & Exception Handling

5. Out of Scope

- Data Warehouse Star Schema.
- Data Profiling Report.
- Power BI Dashboard.
- Data Lineage and Metadata Repository.
- Final Project Report.

6. Stakeholders

Manufacturing Team

Quality Assurance Team

Laboratory Team

Clinical Research Team

Regulatory Affairs Team

Supply Chain Team

Executive Management

Data Engineering Team

Analytics Team

7. Project Team Members

Alfred - Dataset collection, Architecture Diagram, Product Backlog, Sprint Backlog.

Sanika - Dataset collection, Create Git Repo, Repo Structure, BRD.

Thomas - Dataset collection, Identify Stakeholders, BRD.

8. Sprint Deliverables

SPRINT 0 DELIVERABLES

- Business Requirement Document (BRD)
- Project Charter
- Solution Architecture
- GitHub Repository
- Product Backlog
- Sprint Backlog

SPRINT 1 DELIVERABLES

- Source Inventory
- Data Dictionary
- Pentaho ETL Pipelines
- PostgreSQL Staging Database
- ETL Logging and Exception Handling

9. Technology Stack

ETL: Pentaho Data Integration (Spoon)

Database: PostgreSQL

Programming: Python (Pandas)

Version Control: Git & GitHub

Reporting (Future Sprint): Power BI

10. High-Level Architecture

Enterprise Data Sources → Pentaho Data Ingestion → PostgreSQL Staging Database →
Cleaned & Validated Data (Ready for Sprint 2 Warehouse).

11. Success Criteria (Sprint 0 & Sprint 1)

- All pharmaceutical data sources identified.
- Source Inventory and Data Dictionary completed.
- ETL pipelines ingest datasets successfully.
- Raw data loaded into PostgreSQL staging tables.

- ETL logging captures successful and failed records.
- GitHub repository contains Sprint 0 and Sprint 1 deliverables.

12. Risks and Mitigation

Missing values → Validation rules.

Duplicate records → Deduplication during ETL.

Different formats → Pentaho transformation rules.

ETL failures → Logging and exception handling.

13. Assumptions

Sample pharmaceutical datasets are available. Pentaho, PostgreSQL, Python, and GitHub are installed and accessible.

14. Constraints

The project is an academic enterprise data engineering project using publicly available pharmaceutical datasets. Data quality depends on the source datasets.

15. Approval Matrix

- Organization: ABC Pharmaceuticals Ltd.
- Prepared By: Alfred, Sanika, Thomas
- Reviewed By: Dr. M Renukadevi Ma'am
- Status: Sprint 0 & Sprint 1 Submission

16. Conclusion

This Project Charter authorizes only Sprint 0 and Sprint 1 activities for the Enterprise Pharmaceutical Data Platform and provides the roadmap for project initiation, architecture design, data source discovery, ETL ingestion, PostgreSQL staging, and GitHub collaboration.